Fiscal Year:	FY 2014	**Task Last Updated:**	FY 01/15/2014
PI Name:	Volz, Martin Ph.D.		
Project Title:	Reduction of Defects in Germanium-Silicon (RDGS)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	256-544-5078
Organization Name:	NASA Marshall Space Flight Center		
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City:	Huntsville	**State:**	AL
Zip Code:	35812	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-02
Start Date:	03/06/2008	**End Date:**	03/05/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: end date change per P. Downey/MSFC (Ed., 12/21/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mazuruk, Konstantin (University of Alabama Huntsville) Croell, Arne (University of Freiburg)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Original Principal Investigator was Dr. Frank Szofran, Feb 1998-Jan 2004; project continues with Dr. Martin Volz. It is well established that crystals grown without contact with a container have far superior quality to otherwise similar crystals grown in direct contact with a container. In addition to float-zone processing, detached-Bridgman growth is often cited as a promising tool to improve crystal quality, without the limitations of float zoning. In detached growth, most of the melt has contact with the crucible wall, but the crystal is separated from the wall by a small gap, typically 10-100 microns. A liquid meniscus extends between the top edge of the growing crystal and the crucible wall. Detached growth has been found to occur quite often during microgravity experiments and considerable improvements of crystal quality have been reported for those cases. However, no thorough understanding of the process or quantitative assessment of the quality improvements existed prior to the investigation. This project will determine the means to reproducibly grow Ge-Si alloys in the detached mode. Ge-Si alloys are of technological interest because with lower defect density materials it would be possible to make thermal neutron and x-ray optics, faster devices than with silicon, higher efficiency solar cells, and higher thermopower devices. Although these alloys are model materials in the sense that these results should be applicable to many other materials, the major technological significance is that the same production equipment that is currently used for silicon device manufacture can also be used for germanium-silicon devices. This means that using germanium-silicon instead of some other material to achieve improved device performance will save an immense amount of money in development and purchase of production equipment. Several tasks have been completed or are planned as part of the investigation: * measure the relevant material parameters such as contact angle, growth angle, surface tension, and wetting behavior of the GeSi-melt on potential crucible materials; * determine the mechanism of detached growth including the role of convection; * quantitatively determine the differences in defects and impurities among normal Bridgman, detached Bridgman, and float zone (FZ) growth; * investigate the influence of defined azimuthal or meridional flow due to rotating magnetic fields on the characteristics of detached growth; * control time-dependent Marangoni convection in the case of FZ-growth by the use of a rotating magnetic field to examine the influence on the curvature of the solid-liquid interface and the heat and mass transport; and * grow high quality GeSi single crystals with Si concentrations up to 10 %. Determine the necessary conditions for reproducibly achieving detachment of Ge-Si alloys in the Bridgman growth method, and quantitatively compare the defect structures and impurity levels of detached Bridgman, normal Bridgman, and floating-zone grown materials.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research will increase our understanding of the mechanisms which lead to crystal detachment and enhance our ability to control detachment both in microgravity and on Earth. Control of detachment will enable the growth of relatively large diameter crystals with orders of magnitude fewer defects than can be achieved with standard growth techniques. The improvement in material quality may lead to better performance for a variety of electronic devices.
Task Progress:	The current plan is for the flight experiments to utilize three distinct ampoule designs. The different designs are required to accommodate the variation in processing parameters in the flight experiment matrix. These parameters include ampoule material, sample material, and internal ampoule pressure. Laboratory experiments have been conducted to assess and optimize the ampoule designs. These experiments have been conducted using a custom 7-zone Bridgman laboratory furnace. Meetings have been held with engineers working on the Materials Science Research Rack (MSRR) Low Gradient Furnace (LGF) cartridge development. The purpose of these meetings has been to develop a design for the internal layout of the cartridge assembly that will meet both the engineering and science requirements. An Interim Design Review (IDR) of the integrated cartridge and ampoule assembly is expected to occur in 2014. Our study of the static stability of menisci in detached Bridgman growth has been completed. The Young-Laplace equation was used to calculate detached states but the only states that are physically realizable are those which are statically stable. Stable states are those for which the potential energy is at a minimum. The meniscus stability analysis consisted of determining the sign of the second-order correction to the energy for any admissible meniscus perturbations. Axisymmetric as well as non-axisymmetric perturbations were investigated. Both rough and microscopically smooth crucible surfaces, corresponding to pinned and unpinned states, were also considered. Pinning extends the range over which menisci are stable. For Bond numbers equal to zero, corresponding to zero gravity conditions, all menisci are statically stable. The study was published in Physics of Fluids 25, 094106 (2013).
Bibliography Type:	Description: (Last Updated: 07/09/2021)
Abstracts for Journals and Proceedings	Volz MP, Mazuruk K. "Crystal Shape Evolution in Detached Bridgman Growth." Presented at the 17th International Conference on Crystal Growth and Epitaxy (ICCGE-17), Warsaw, Poland, August 11-16, 2013. 17th International Conference on Crystal Growth and Epitaxy (ICCGE-17), Warsaw, Poland, August 11-16, 2013. Abstracts, p. 487-488. , Aug-2013
Abstracts for Journals and Proceedings	Mazuruk K, Volz MP. "Stability of Menisci in Detached Bridgman Growth." Presented at the 17th International Conference on Crystal Growth and Epitaxy (ICCGE-17), Warsaw, Poland, August 11-16, 2013. 17th International Conference on Crystal Growth and Epitaxy (ICCGE-17), Warsaw, Poland, August 11-16, 2013. Abstracts, p. 487. , Aug-2013

Abstracts for Journals and Proceedings	Volz MP, Mazuruk K. "Evolution of the Shape of Detached GeSi Crystals in Microgravity." Presented at the 29th American Society for Gravitational and Space Research and 5th International Symposium for Physical Sciences in Space, Orlando, Florida, November 3-8, 2013. 9th American Society for Gravitational and Space Research and 5th International Symposium for Physical Sciences in Space, Orlando, Florida, November 3-8, 2013. Program and abstracts, p. 59. , Nov-2013
Articles in Peer-reviewed Journals	Mazuruk K, Volz MP. "Static stability of menisci in detached Bridgman growth." Physics of Fluids. 2013 Sep;25(9):094106. http://dx.doi.org/10.1063/1.4821849 , Sep-2013